KARIS

Kitting Automation Robots (in) Industrial Settings

Vision: Kim Svedberg, Gustav Andrée Fröjd, Gustav Samuelson,

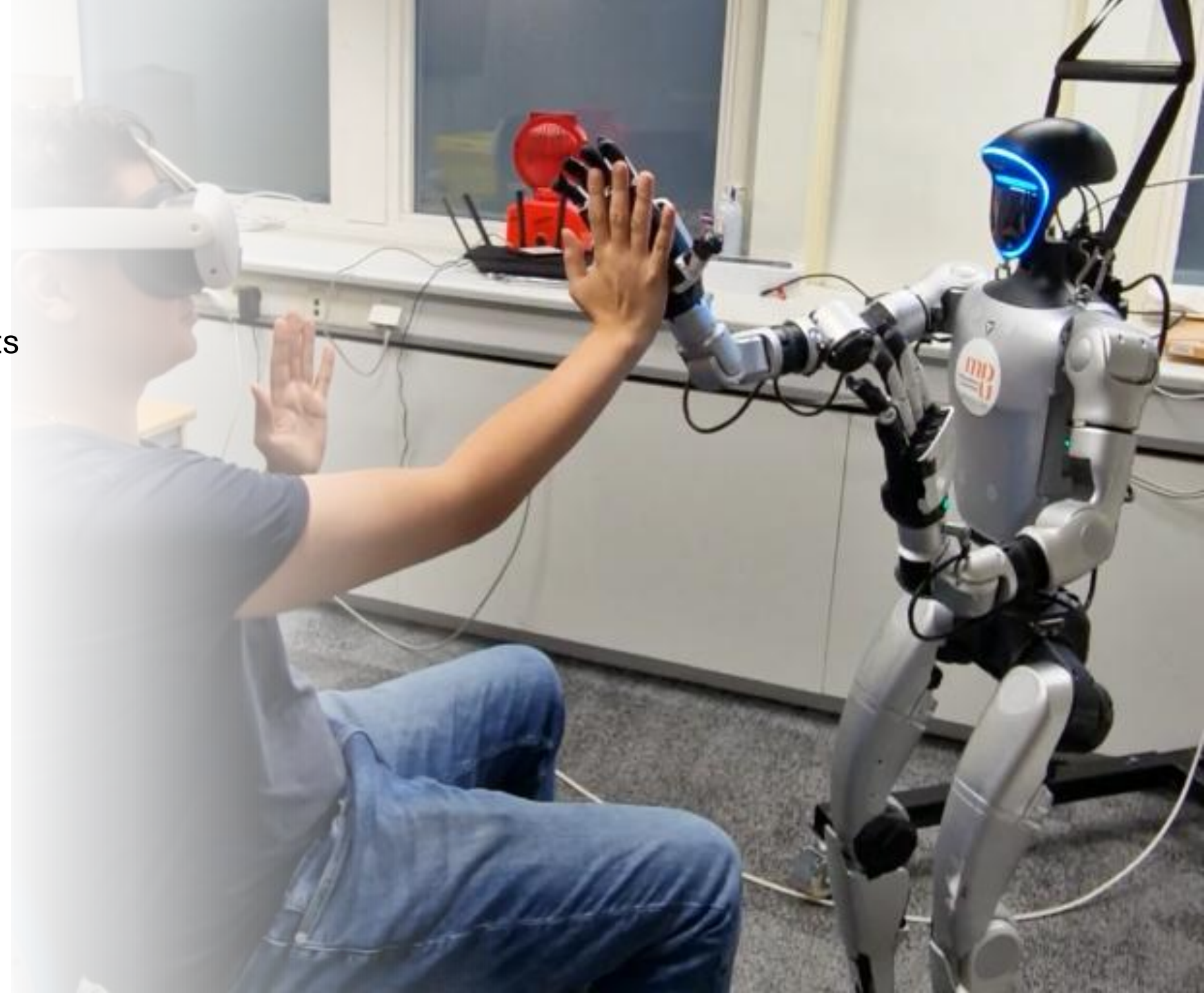
Motion: Joni Ojala, Whilliam Borglund, Oskar Frej

Stakeholders: Lars Hanson (Volvo), Ludvig Svedlund (MDU)

Purpose

The aim of this project is to evaluate the feasibility of integrating humanoid robots into existing industrial environments,

enabling them to perform tasks traditionally carried out by human operators without requiring modifications to the workplace.



Background

Unitree G1 is equipped with:

Stereo-vision camera in combination with a 3D LiDAR.

FTP Inspire hands for precision tasks ~800g each.

Two PC’s internally, Locomotion (PC1), Development (PC2). They run ROS-Foxy

Key requirements for the project:

ROS-based communication.

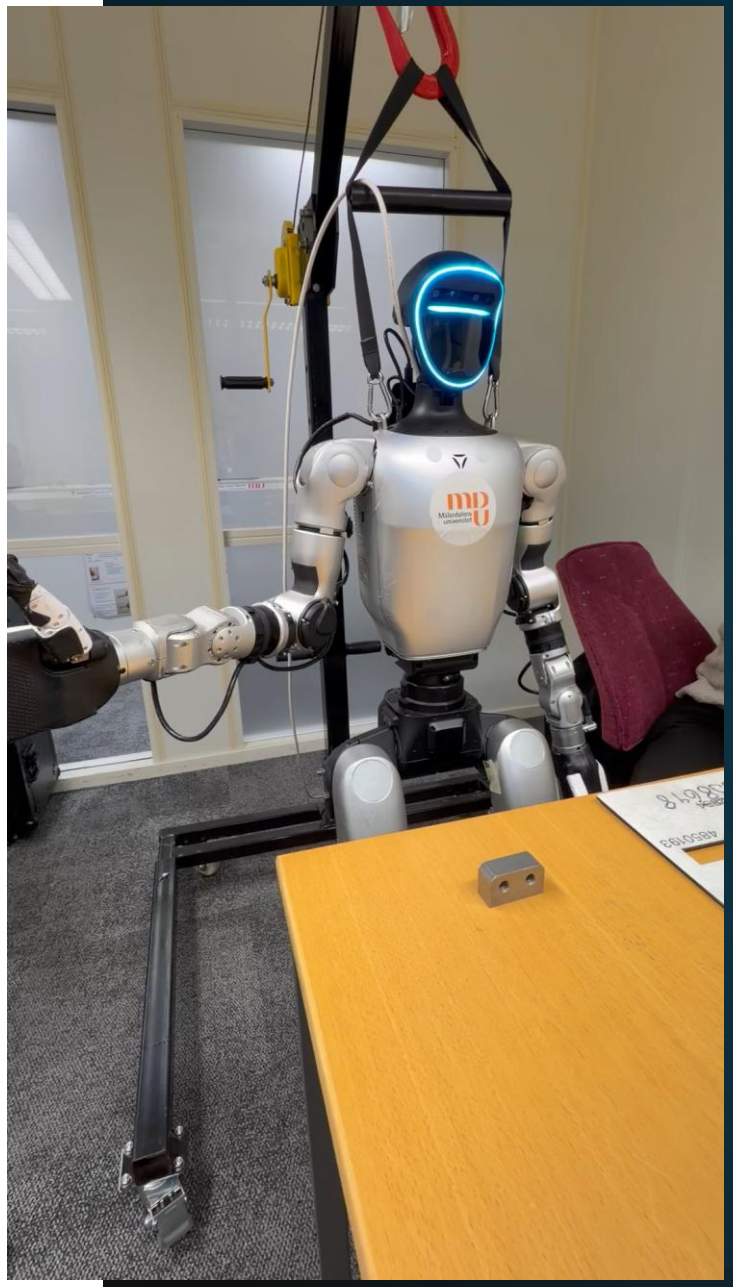
Must pick, place and manipulate kitted objects without dropping them, in a timeframe comparable to a human operator.

Must not damage itself, other items, structures, or living entities. No existing hardware may be modified without written permission.



Expected outcome

- Identify parts
- Estimate pose
- Move arms and hands
- Pick and place



Teamwork

- Vision

  Kim Svedberg

  Gustav Andrée Fröjd

  Gustav Samuelson

- Motion

  Joni Ojala

  Oskar Frej

  Whilliam Borglund



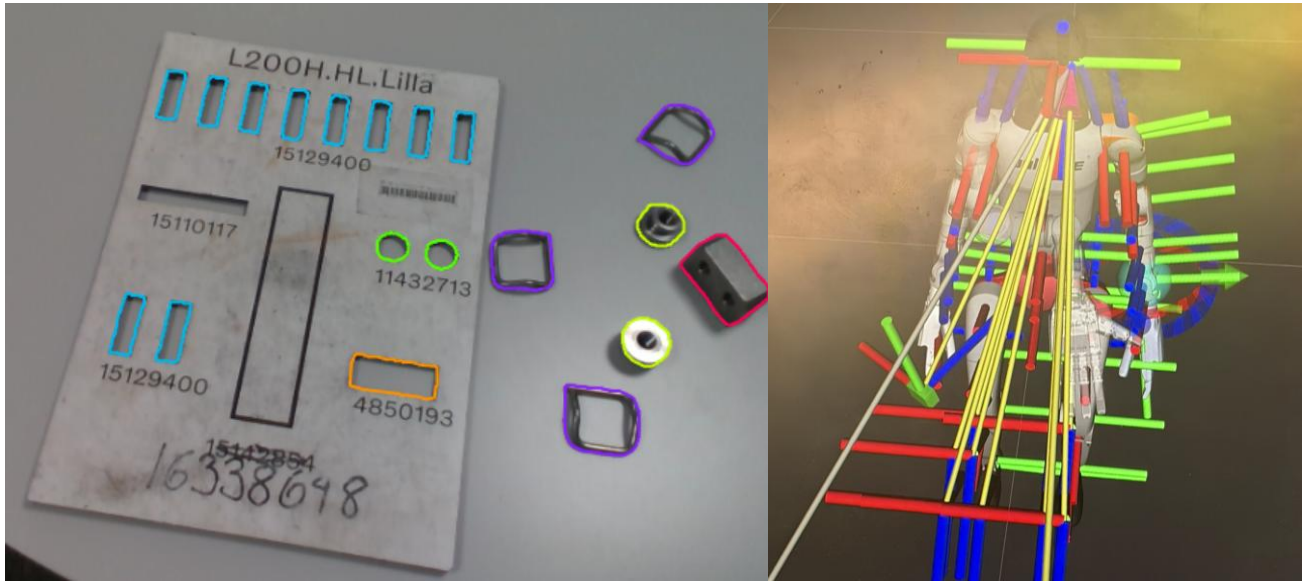
Daily standup meetings 14.30



Weekly meeting with supervisor
11.00 Thursdays

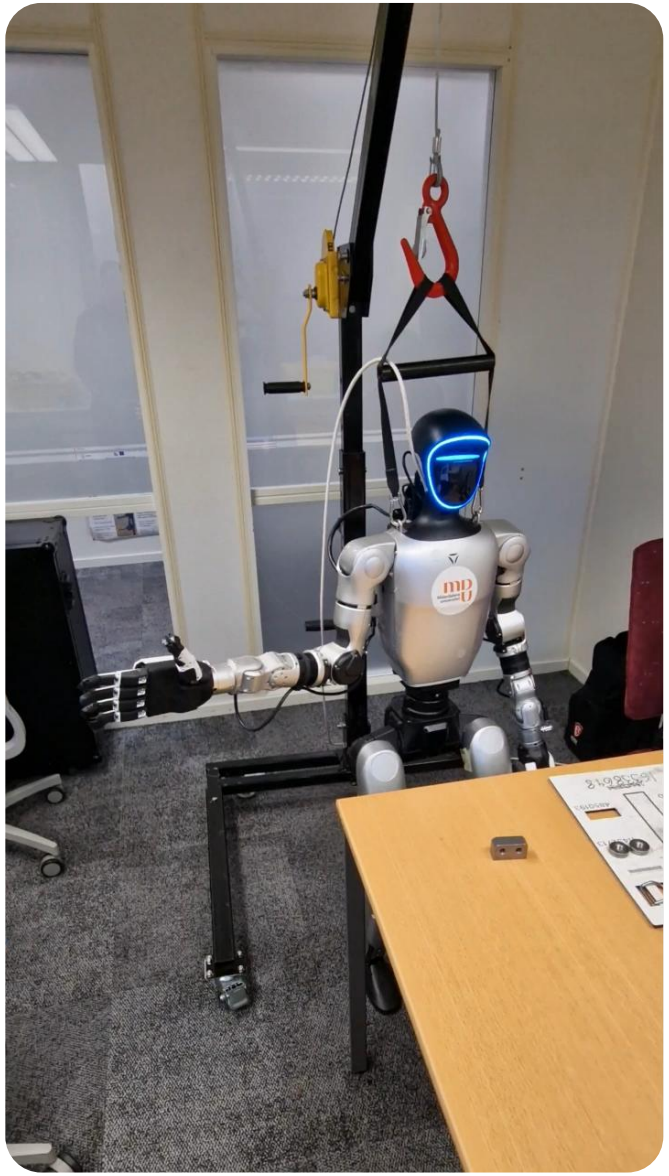
Progress (Vision)

- Collect training data
- Train YOLO and RF-DETR models
- Align the color and depth image
- FoundationPose for orientation (one class)
- Send images and poses using ROS topics

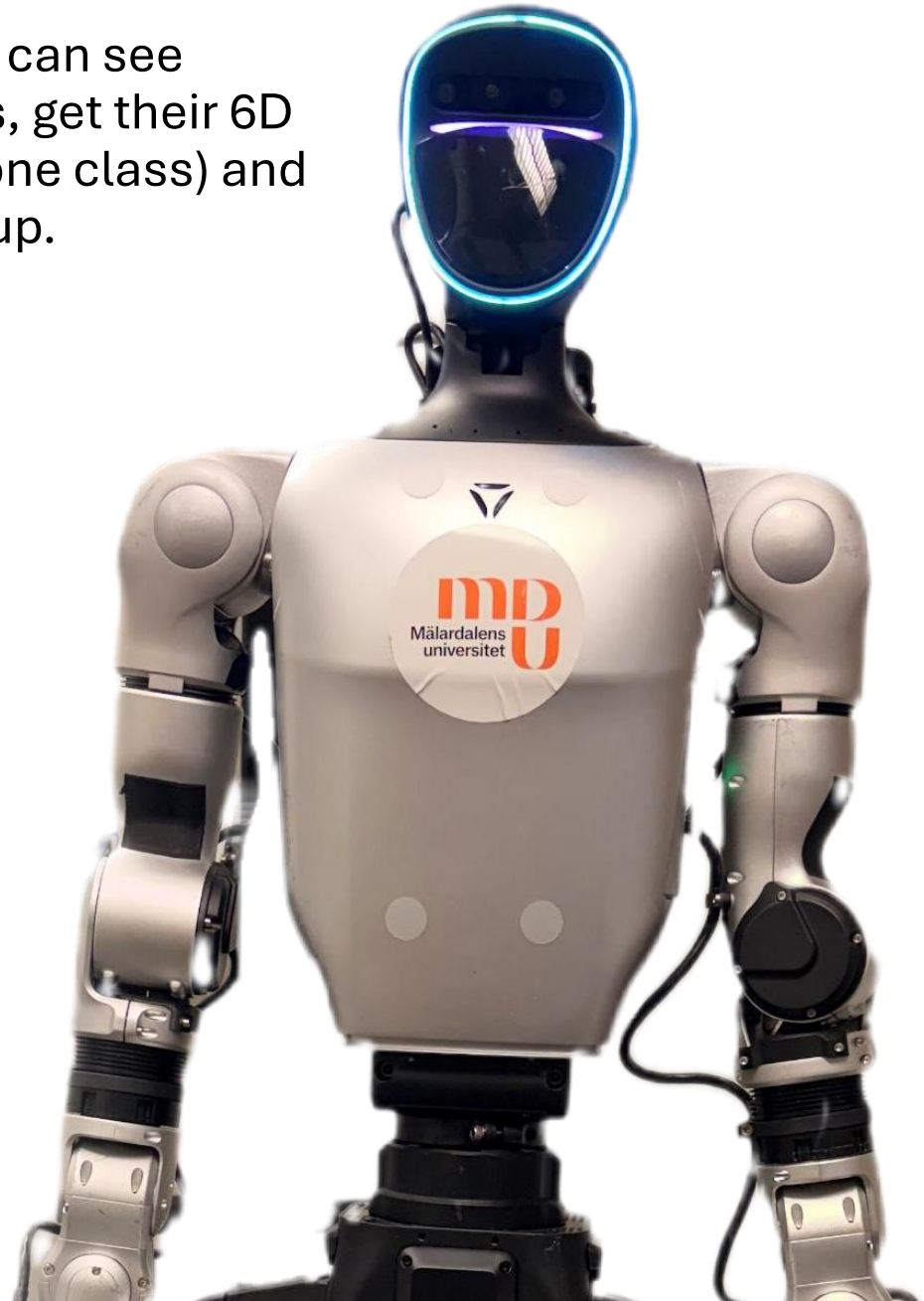




Progress (Motion)

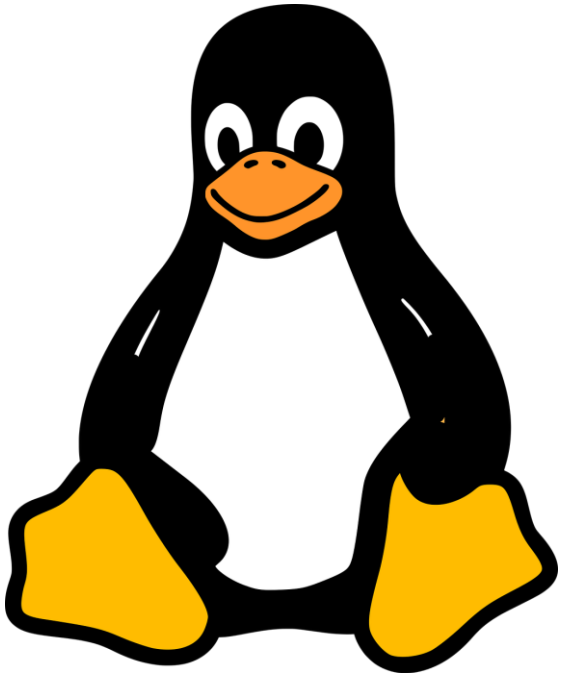


The G1 can see objects, get their 6D pose (one class) and pick it up.

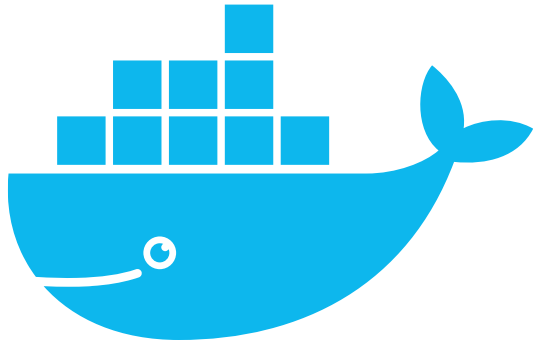


Problems (Software)

- Linux (solved)
  - Versions
- ROS (solved)
  - Distros
- Docker (solution)
  - No knowledge - solved the problems
- DDS/Data communication/SSH (solved and current problems)
  - G1, Linux, Windows, Wifi, Ethernet.

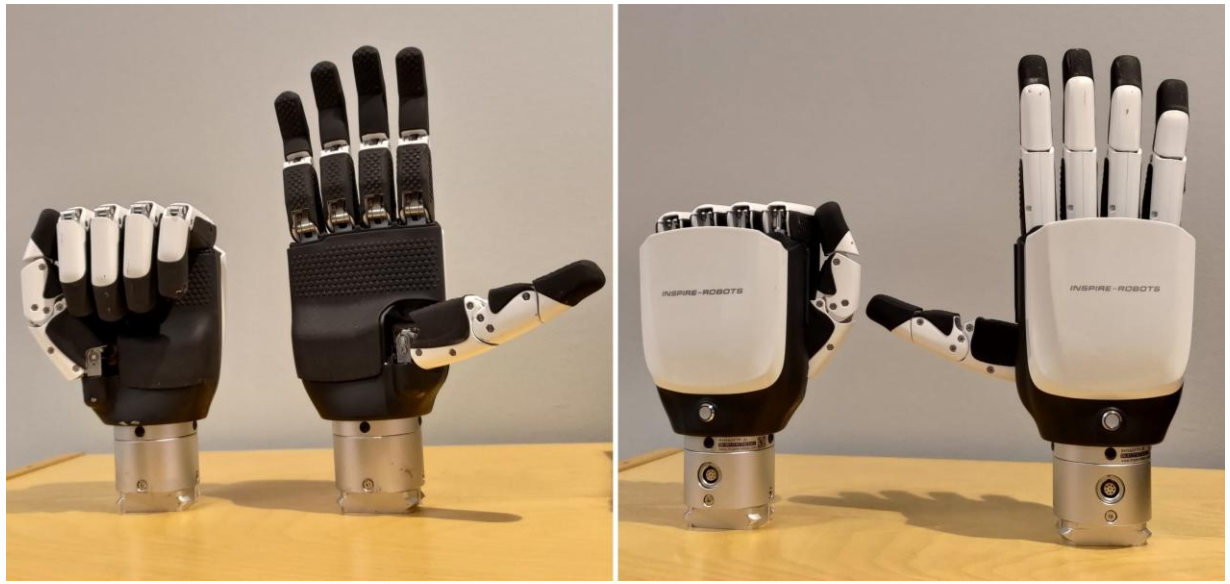






Problems (Hardware)

- Battery (not solved)
- GPU access (solved and currently working on)
- Better hands (not solved)





Future Challenges

- Inaccurate pose of object
- Non-ideal environment
- Inaccurate robot movement
- Move everything